2-ACRE MORINGA COMMERCIAL ORCHARD — IRRIGATION ENGINEERING MASTER BLUEPRINT

HYDRAULIC DISTRIBUTION NETWORK • OVERHEAD DOUBLE-TEE RISERS • INLINE & ONLINE DRIP LAYOUT

STANDARDS: IS 12786 (DRIP LATERALS) | IS 4985 (PVC PRESSURE PIPELINES) | DESIGN PRESSURE: 2.5 KG/CM²

CENTRAL ANNOTATION CHANNEL (ZERO-OVERLAP)
165'-0" (PLOT B WIDTH)
PLOT B: BHAGYA MORINGA (1 ACRE | 7ft x 8ft GRID | 750 TREES)
32 ROWS (WESTWARD ONLY) • 16MM PLAIN LATERALS • 24 PUNCHED TUNNING DRIPPERS/ROW (ITEM #2)
40MM FLUSH VALVE (NORTH TERMINAL)
ELEVATION: -2'-0" IN SURFACE SLEEVE
COVERS TOP 4 ROWS (ROWS #29-32)
RISER B4 (OFFSET: 248'-0" NORTH)
2" MAINLINE TERMINATES HERE (2" END CAP)
OVERHEAD 2" TEE + 2x 40MM BALL VALVES
ONLY 1.25" SUBLINE CONTINUES 32 FT NORTH
RISER B3 (OFFSET: 184'-0" NORTH)
4FT DOUBLE-TEE + 2x 40MM BALL VALVES
1.25" SUBLINES: 32' SOUTH & 32' NORTH
FEEDS 8 ROWS (ROWS #17-24 | 192 TREES)
256'-0" ACTIVE HEIGHT
32 ROWS @ 8'-0" SPACING
RISER B2 (OFFSET: 120'-0" NORTH)
4FT DOUBLE-TEE + 2x 40MM BALL VALVES
1.25" SUBLINES: 32' SOUTH & 32' NORTH
FEEDS 8 ROWS (ROWS #9-16 | 192 TREES)
145'-0" CENTER MAINLINE
24 TRENCHES @ 5' SPACING
120 FT ACTIVE SUBLINE
RISER B1 (OFFSET: 56'-0" NORTH)
4FT DOUBLE-TEE + 2x 40MM BALL VALVES
SOUTH ARM EXTENDS 32' TO PLOT B START (24')
FEEDS 8 ROWS (ROWS #1-8 | 192 TREES)
PLOT B START BOUNDARY (24'-0" FROM TEE B)
40MM FLUSH VALVE (SOUTH ARM TERMINAL)
ROW #1 STARTS HERE (NO ROWS BELOW THIS LINE)
24' BARE GROUND RUN
(NO PLANTS / NO ROWS)
2" END CAP
TEE B: PLOT B ENTRY
COORD: (285', 100')
2" TEE WITH WEST CAP
390'-0" (PLOT A OVERALL TRENCH WIDTH: 195' W + 195' E)
PLOT A: PKM-1 ANNUAL MORINGA (1 ACRE | 3ft x 5ft HIGH-DENSITY | 3,000 PLANTS)
24 ACTIVE TRENCHES @ 5' SPACING • 16x4x40 INLINE DRIP LATERALS • 50x16x2-WAY TAKE-OFF COLLARS (V-CONNECTORS)
16x4x40 INLINE LA
390 FT TOTAL TREN
195 FT WEST + 195
16MM ENDCAPS ON
FEEDS 6 TRENCHES
RISER A3 (OFFSET
OVERHEAD 2" TEE
1.5" SUBLINES: 15
FEEDS 6 TRENCHES
RISER A2 (OFFSET
OVERHEAD 2" TEE
1.5" SUBLINES: 15
FEEDS 6 TRENCHES
RISER A1 (OFFSET
OVERHEAD 2" TEE
1.5" SUBLINES: 15
FEEDS 6 TRENCHES
215'-0" 2" MAINLINE RUN (UNDER PLOT A & ROADWAY BUFFER)
TRUE PIPE LENGTH: 215 FT | TEE A (500', 100') TO TEE B (285', 100')
90'-0" 2" PVC MAINLINE (63MM)
2" END CAP
TEE A: PLOT A ENTRY

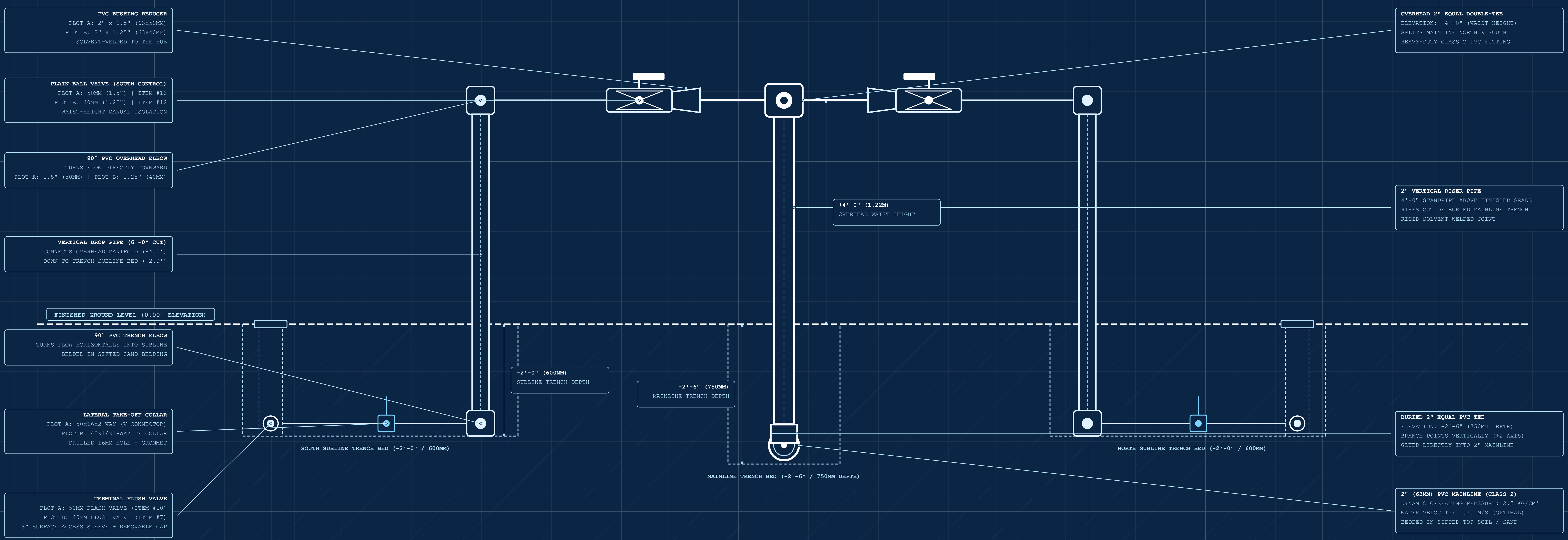
PROJECT & LOCATION: **2-ACRE MORINGA COMMERCIAL ORCHARD DRIP SYSTEM**		
DRAWING TITLE: **MASTER FIELD BLUEPRINT**	ZONE BREAKDOWN: **ZONE 1: 5HP SOLAR BOREWELL (2" MAINLINE)** ZONE 2: PLOT A (PKM-1, 24 TRENCHES, 3000P)	ZONE 3 (PLOT B SPEC): **BHAGYA (7X8 FT, 32 ROWS, 750P)** MAINLINE TERMINATES AT RISER B4
HYDRAULIC CRITERIA: **DESIGN VELOCITY: < 1.5 M/S** HEAD LOSS: < 2.0 M / 100M	FITTINGS SCHEDULE: **16x PLAIN BALL VALVES** 16x FLUSH VALVES (SURFACE)	EMITTER SPECIFICATION: **PLOT A: 4 LPH INLINE @ 40CM** PLOT B: TUNNING DRIPPER @ 7FT

SCALE: **1 FT = 5 PX** (1:60 ON A0 / ARCH E)	DATE / STATUS: **OCTOBER 2026** APPROVED FOR FIELD	CAD STANDARD: **IS 12786 / 4985** CLASSIC BLUEPRINT	SHEET NUMBER: **SHEET 1 OF 2**

4-FOOT OVERHEAD DOUBLE-TEE RISER MANIFOLD — DETAIL CROSS-SECTION

SIDE-ELEVATION LOOKING EAST • WAIST-HEIGHT VALVE CONTROLS • TRENCH DROP PROFILES

APPLIES TO RISERS A1-A4 (50MM BALL VALVES) & RISERS B1-B4 (40MM BALL VALVES) | SCALE: 1 FT = 60 PX


FINISHED GROUND LEVEL (0.00' ELEVATION)
+4'-0" (1.22M)
OVERHEAD WAIST HEIGHT
-2'-0" (600MM)
SUBLINE TRENCH DEPTH
-2'-6" (750MM)
MAINLINE TRENCH DEPTH
SOUTH SUBLINE TRENCH BED (-2'-0" / 600MM)
MAINLINE TRENCH BED (-2'-6" / 750MM DEPTH)
NORTH SUBLINE TRENCH BED (-2'-0" / 600MM)

PVC BUSHING REDUCER
PLOT A: 2" x 1.5" (63x50MM)
PLOT B: 2" x 1.25" (63x40MM)
SOLVENT-WELDED TO TEE HUB

PLAIN BALL VALVE (SOUTH CONTROL)
PLOT A: 50MM (1.5") | ITEM #13
PLOT B: 40MM (1.25") | ITEM #12
WAIST-HEIGHT MANUAL ISOLATION

90° PVC OVERHEAD ELBOW
TURNS FLOW DIRECTLY DOWNWARD
PLOT A: 1.5" (50MM) | PLOT B: 1.25" (40MM)

VERTICAL DROP PIPE (6'-0" CUT)
CONNECTS OVERHEAD MANIFOLD (+4.0')
DOWN TO TRENCH SUBLINE BED (-2.0')

90° PVC TRENCH ELBOW
TURNS FLOW HORIZONTALLY INTO SUBLINE
BEDDED IN SIFTED SAND BEDDING

LATERAL TAKE-OFF COLLAR
PLOT A: 50x16x2-WAY (V-CONNECTOR)
PLOT B: 40x16x1-WAY TF COLLAR
DRILLED 16MM HOLE + GROMMET

TERMINAL FLUSH VALVE
PLOT A: 50MM FLASH VALVE (ITEM #10)
PLOT B: 40MM FLUSH VALVE (ITEM #7)
8" SURFACE ACCESS SLEEVE + REMOVABLE CAP

OVERHEAD 2" EQUAL DOUBLE-TEE
ELEVATION: +4'-0" (WAIST HEIGHT)
SPLITS MAINLINE NORTH & SOUTH
HEAVY-DUTY CLASS 2 PVC FITTING

2" VERTICAL RISER PIPE
4'-0" STANDPIPE ABOVE FINISHED GRADE
RISES OUT OF BURIED MAINLINE TRENCH
RIGID SOLVENT-WELDED JOINT

BURIED 2" EQUAL PVC TEE
ELEVATION: -2'-6" (750MM DEPTH)
BRANCH POINTS VERTICALLY (+Z AXIS)
GLUED DIRECTLY INTO 2" MAINLINE

2" (63MM) PVC MAINLINE (CLASS 2)
DYNAMIC OPERATING PRESSURE: 2.5 KG/CM²
WATER VELOCITY: 1.15 M/S (OPTIMAL)
BEDDED IN SIFTED TOP SOIL / SAND

PROJECT & LOCATION:
2-ACRE MORINGA COMMERCIAL ORCHARD DRIP SYSTEM

DRAWING TITLE: **OVERHEAD RISER DETAIL**	PLOT A CONFIGURATION: **4 STATIONS (A1-A4 @ 30' CTRS)** 2x 50MM VALVES + 15' SUBLINES	PLOT B CONFIGURATION: **4 STATIONS (B1-B4 @ 64' CTRS)** 2x 40MM VALVES + 32' SUBLINES
OPERATIONAL HEIGHT: **+4'-0" (1.22M) WAIST LEVEL** ERGONOMIC ROTATION ACCESS	PLUMBING SPECIFICATION: **HEAVY SOLVENT CEMENT** 24-HR HYDROSTATIC CURE	TRENCH DRAINAGE: **SURFACE FLUSH SLEEVES** EASY MAINTENANCE ACCESS

SCALE: **1 FT = 120 PX** (1:10 DETAIL SCALE)	DATE / STATUS: **OCTOBER 2026** APPROVED FOR FIELD	CAD STANDARD: **IS 12786 / 4985** CROSS-SECTION DETAIL	SHEET NUMBER: **SHEET 2 OF 2**